

Minimisation II: Brzozowski Strikes Back

Nelma Moreira, Rogério Reis, **Gonçalo Teixeira**

DCC & CMUP
Faculty of Sciences University of Porto

September 3, 2026

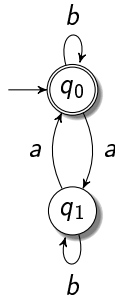
NOVA FCT, Lisbon, INForum 2026

Table of Contents

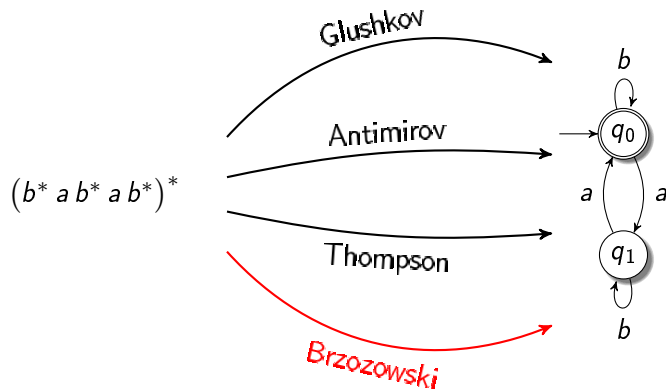
1. Regular Expressions to DFAs
2. Head Normal Form
3. Minimisation
4. Conclusion

Two Views Of Regular Languages

$$(b^* a b^* a b^*)^*$$



Two Views Of Regular Languages



Regular Expressions

$$r \rightarrow \emptyset \mid \varepsilon \mid \sigma \in \Sigma \mid r + r \mid r \cdot r \mid r^*$$

DFA

A deterministic finite automaton is a 5-tuple $A = (Q, \Sigma, \delta, q_0, F)$, where

- Q is a finite set of states
- Σ is a finite non-empty alphabet
- $\delta : Q \times \Sigma \rightarrow Q$ is a transition function
- $q_0 \in Q$ is the initial state
- $F \subseteq Q$ is the set of final states

The language recognized by A is $\mathcal{L}(A) = \{w \mid \delta(q_0, w) \in F\}$.

A DFA is minimal if it has the least number of states of all DFAs recognizing the same language.

Brzozowski Derivatives

Nullability $\varepsilon : RE \rightarrow \{\emptyset, \varepsilon\}$:

$$\varepsilon(\emptyset) = \emptyset, \quad \varepsilon(\varepsilon) = \varepsilon, \quad \varepsilon(\sigma) = \emptyset$$

$$\varepsilon(r + s) = \varepsilon(r) + \varepsilon(s)$$

$$\varepsilon(r \cdot s) = \varepsilon(r) \cdot \varepsilon(s), \quad \varepsilon(r^*) = \varepsilon$$

Derivative $d : \Sigma \times RE \rightarrow RE$:

$$d_\sigma(\emptyset) = \emptyset, \quad d_\sigma(\sigma') = [\sigma = \sigma']$$

$$d_\sigma(r + s) = d_\sigma(r) + d_\sigma(s), \quad d_\sigma(r^*) = d_\sigma(r)r^*$$

$$d_\sigma(r \cdot s) = d_\sigma(r)s + \varepsilon(r)d_\sigma(s)$$

Extend to words:

$$d_\varepsilon(r) = r, \quad d_{\sigma w}(r) = d_w(d_\sigma(r))$$

Key Properties:

- $\varepsilon(r) = \varepsilon$ iff $\varepsilon \in \mathcal{L}(r)$
- $\mathcal{L}(d_w(r)) = w^{-1}\mathcal{L}(r)$
- $\mathcal{L}(r) = \mathcal{L}(\varepsilon(r) + \sum_{\sigma \in \Sigma} d_\sigma(r))$
- The set of derivatives $D(r)$ is finite modulo *ACI*

Brzowski Automaton

Brzowski Automaton

Given a regular expression r , the Brzowski automaton $Brz(r)$ is the 5-tuple $(D(r), \Sigma, \delta, [r], F)$, where

$$\delta([r], \sigma) = [d_\sigma(r)], \quad F = \{[r] \mid \varepsilon(r) = \varepsilon\}$$

Brzowski Automaton

Brzowski Automaton

Given a regular expression r , the Brzowski automaton $Brz(r)$ is the 5-tuple $(D(r), \Sigma, \delta, [r], F)$, where

$$\delta([r], \sigma) = [d_\sigma(r)], \quad F = \{[r] \mid \varepsilon(r) = \varepsilon\}$$

Question:

For which classes of regular expressions r is $Brz(r)$ minimal?

A complete characterization in terms of necessary and sufficient conditions seems out of reach, but...

$$r \rightarrow \emptyset \mid \varepsilon \mid \sigma \in \Sigma \mid r + r \mid r \cdot r$$

$$r \rightarrow \emptyset \mid \varepsilon \mid \sigma \in \Sigma \mid r + r \mid r \cdot r$$

- DFAs are acyclic (ADFA)
- Rank $rk(q)$: length of the longest word accepted by q
- Equivalent states have the same rank
- Transitions decrease rank

$$r \rightarrow \emptyset \mid \varepsilon \mid \sigma \in \Sigma \mid r + r \mid r \cdot r$$

- DFAs are acyclic (ADFA)
- Rank $rk(q)$: length of the longest word accepted by q
- Equivalent states have the same rank
- Transitions decrease rank

Minimality of an ADFA (Revuz)

An ADFA A is minimal iff for all distinct $p, q \in Q$:

$$(p \in F \dot{\vee} q \in F) \vee (\exists \sigma \in \Sigma) \delta(p, \sigma) \neq \delta(q, \sigma)$$

$$r \rightarrow \emptyset \mid \varepsilon \mid \sigma \in \Sigma \mid r + r \mid r \cdot r$$

- DFAs are acyclic (ADFA)
- Rank $rk(q)$: length of the longest word accepted by q
- Equivalent states have the same rank
- Transitions decrease rank

Minimality of an ADFA (Revuz)

An ADFA A is minimal iff for all distinct $p, q \in Q$:

$$(\varepsilon(p) = \varepsilon \vee \varepsilon(q) = \varepsilon) \vee (\exists \sigma \in \Sigma) d_\sigma(p) \not\equiv_{ACI} d_\sigma(q)$$

Table of Contents

1. Regular Expressions to DFAs
2. Head Normal Form
3. Minimisation
4. Conclusion

Head Normal Form

Head normal form (HNF)

A finite regular expression is in HNF if it has the form

$$f + \sum_{\sigma \in \Sigma} \sigma \cdot r_{\sigma}$$

where $f \in \{\emptyset, \varepsilon\}$ and each r_{σ} is itself in HNF.

In particular, $\emptyset$, ε , and each $\sigma \in \Sigma$ are degenerate cases of HNF.

Calculating HNF

$$\text{hnf}(r) = \begin{cases} \emptyset & \text{if } r = \emptyset \\ \varepsilon & \text{if } r = \varepsilon \\ \sigma & \text{if } r = \sigma \in \Sigma \\ \varepsilon(r) + \sum_{\sigma \in \Sigma} \sigma \cdot \text{hnf}(d_{\sigma}(r)) & \text{otherwise} \end{cases}$$

- **Termination:** Since derivatives decrease the size of the largest recognized word there is a decreasing well-founded measure
- **Preservation:** $\mathcal{L}(\text{hnf}(r)) = \mathcal{L}(r)$ by induction and the Fundamental Theorem
- **Form:** $\text{hnf}(r)$ is HNF by induction

Properties of HNF

We can take states of $Brz(r)$ to be in HNF

Lemma (Derivative Closed)

If r is in HNF, then $d_\sigma(r)$ is in HNF

Most importantly (and unlike general regular expressions):

Lemma (Revuz Criteria)

For r, s in HNF:

$$r \not\equiv_{ACI} s \implies (\varepsilon(r) = \varepsilon \dot{\vee} \varepsilon(s) = \varepsilon) \vee (\exists \sigma \in \Sigma) d_\sigma(r) \not\equiv_{ACI} d_\sigma(s)$$

Theorem (Minimality)

Let r be in HNF. Then $Brz(r)$ is minimal.

Let $r \not\equiv_{ACI} s$ in HNF

$$r = f_r + \sum_{\sigma \in \Sigma} \sigma r_\sigma \quad s = f_s + \sum_{\sigma \in \Sigma} \sigma s_\sigma$$

If they are not equivalent, then they differ somewhere:

- If $f_r \neq f_s$:

$$\varepsilon(r) = \varepsilon \dot{\vee} \varepsilon(s) = \varepsilon$$

- If for some $\sigma \in \Sigma$, $r_\sigma \not\equiv_{ACI} s_\sigma$:

$$r_\sigma = d_\sigma(r) \not\equiv_{ACI} d_\sigma(s) = s_\sigma$$

Table of Contents

1. Regular Expressions to DFAs
2. Head Normal Form
3. Minimisation
4. Conclusion

Minimisation

Using the Equational Method and processing the states in increasing rank order:

$$r_{\Omega} = \emptyset, \quad r_q = [q \in F] + \sum_{\sigma \in \Sigma} \sigma r_{\delta(q, \sigma)}$$

Acyclicity means that we don't need Arden's Lemma to solve this system of equations
So the resulting expression $r_A = r_{q_0}$ is in HNF

Minimisation Algorithm

Let A be an ADFA. Then $\text{Brz}(r_A)$ is minimal.

Table of Contents

1. Regular Expressions to DFAs
2. Head Normal Form
3. Minimisation
4. Conclusion

Conclusion & Future Work

Future Work:

- Can the minimality result be extended beyond finite languages / HNF to broader classes?
- Necessary and sufficient conditions on r for $Brz(r)$ to be minimal.
- The average size of HNF expressions and the average state complexity of the resulting minimal ADFAs?

Thank You!

